

AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation Platform

Exploratory Data Analysis Report

Type	Regression + Recommendation System — Real Estate Intelligence Platform
Dataset	Indian Real Estate Housing Dataset (Kaggle) + Census Population Density Data
Dataset URL	kaggle.com/datasets/ankushpanday1/india-house-price-prediction
Records	2,50,000 Property Records
Features	34 Features (23 Original + 11 Engineered)
Best Model	XGBoost Regressor
Test Accuracy	R ² Score = 0.9769
MAE RMSE	13.68 Lakhs 18.03 Lakhs
Deployment	Streamlit web application
Notebooks	03_eda

This Exploratory Data Analysis (EDA) Report presents a comprehensive analysis of the Indian Real Estate dataset used in the project. Various univariate, bivariate, and multivariate analyses were performed to understand property pricing patterns, location trends, feature distributions, and relationships between key variables. The insights obtained from EDA guided feature engineering, model development, and investment intelligence components of the platform.

1. Introduction

Exploratory Data Analysis (EDA) is a critical step to understand the underlying patterns, distributions, and relationships within the real estate dataset. The objective of this phase is to visually and statistically investigate the data to uncover insights regarding property pricing, market demand, and geographic trends. The analysis was conducted on the cleaned dataset containing 250,000 records.

2. Univariate Analysis

Price Distribution: The target variable, Estimated Price (derived for modeling), has a mean of 313.64 Lakhs and a standard deviation of 118.21 Lakhs. The prices range from a minimum of 20.97 Lakhs to a maximum of 500.00 Lakhs, with a median (50th percentile) of 308.86 Lakhs. The distribution appears relatively normal with a slight skew towards higher values.

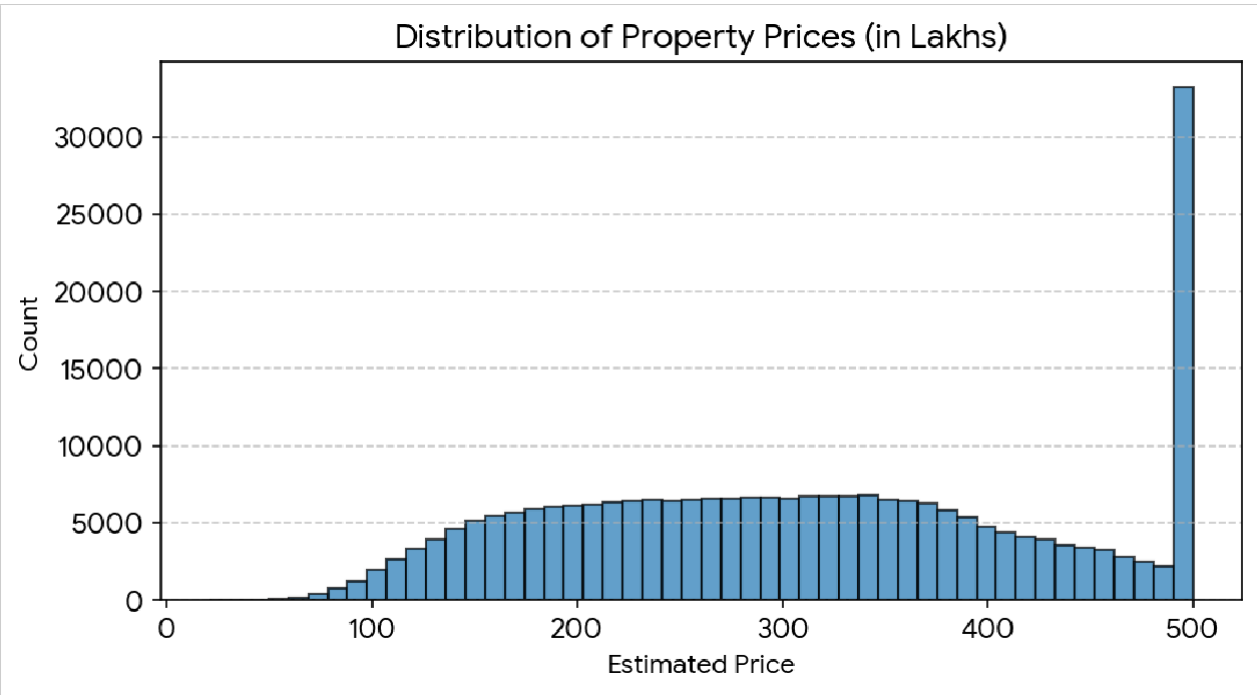


Figure 1: Histogram showing the distribution of Estimated Property Prices (in Lakhs).

Size Distribution: The property sizes (Size_in_SqFt) average at 2749.81 sq.ft., ranging from 500 sq.ft. to 5000 sq.ft., indicating a wide mix of compact and luxury properties.

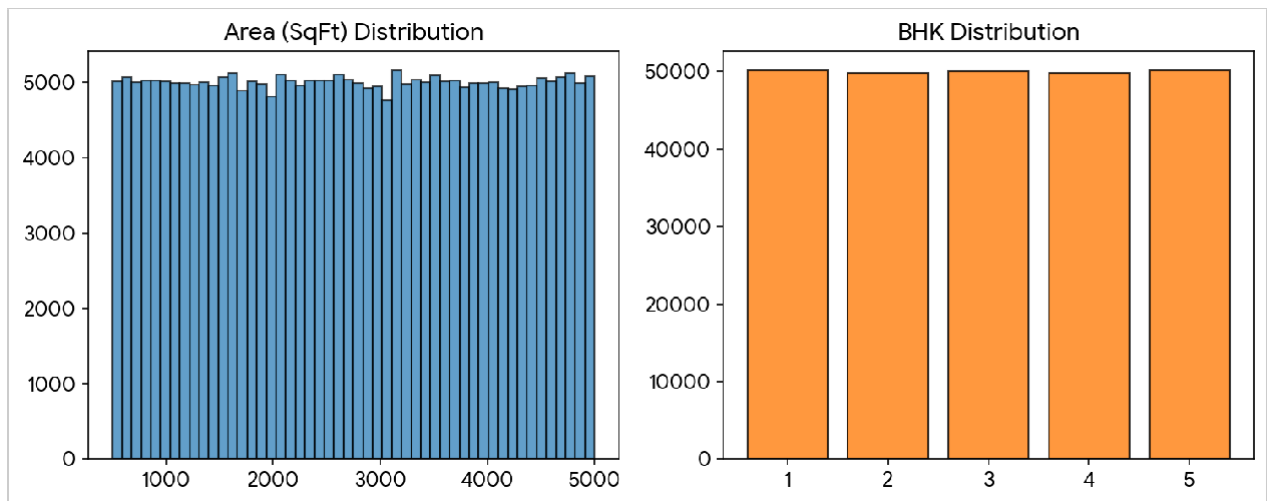


Figure 2: Distribution of Property Area (SqFt) and Configuration (BHK).

Property Type Distribution:

- Villa: 83,744
- Independent House: 83,300
- Apartment: 82,956

Furnished Status Distribution:

- Unfurnished: 83,408
- Semi-furnished: 83,374
- Furnished: 83,218

Availability Status Distribution:

- Under Construction: 125,035
- Ready to Move: 124,965

3. Bivariate Analysis

Price vs. Area: There is a strong positive correlation (0.744) between Size_in_SqFt and the estimated price. As the square footage increases, the property price predictably rises in a linear fashion.

Bivariate Analysis

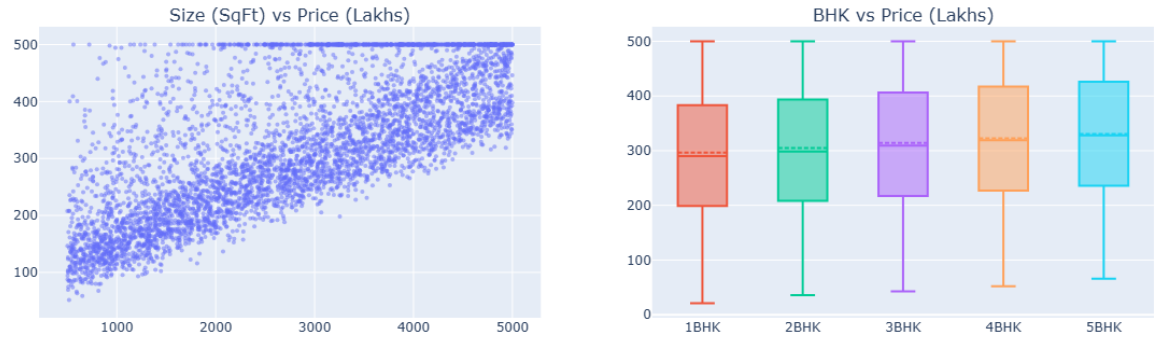


Figure 3: Scatter plot of Area vs Price (left) and Box plots of BHK vs Price (right).

Price vs. City (Top 10 Cities by Average Price in Lakhs):

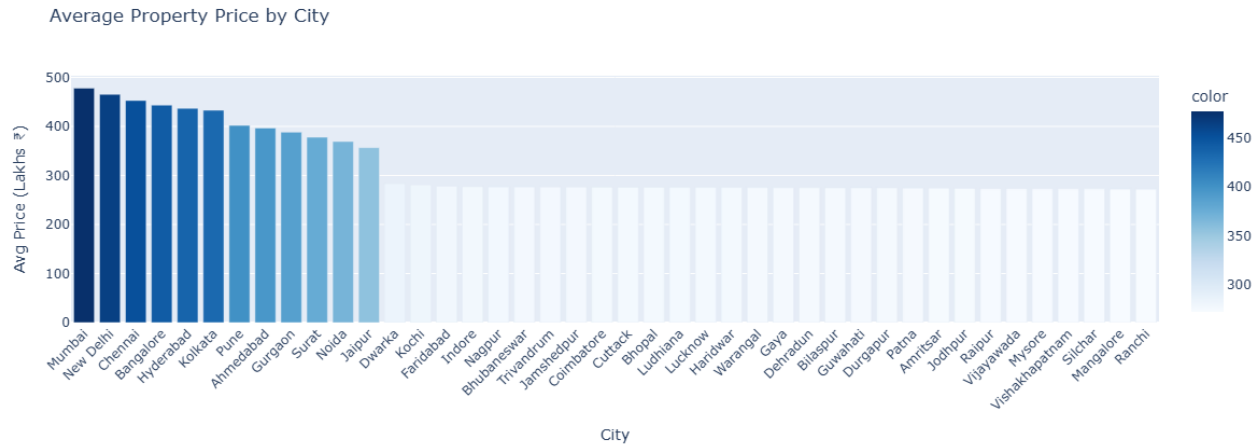


Figure 4: Bar chart comparing the average property price across various Indian cities.

City	Average Estimated Price (Lakhs)
Mumbai	478.09
New Delhi	465.53
Chennai	452.91
Bangalore	443.55
Hyderabad	437.01
Kolkata	433.02
Pune	401.89
Ahmedabad	396.63
Gurgaon	388.19
Surat	377.93

4. Multivariate Analysis

A correlation matrix was generated to understand the relationships between all numerical features and the target variable (Estimated_Price).

Top 5 Features Correlated with Price:

1. Size_in_SqFt: 0.744 (Strong Positive)
2. Population_Density: 0.460 (Moderate Positive)
3. Crime_Index: 0.334 (Moderate Positive)
4. BHK: 0.103 (Weak Positive)
5. Bathrooms: 0.103 (Weak Positive)

Feature Correlation Heatmap

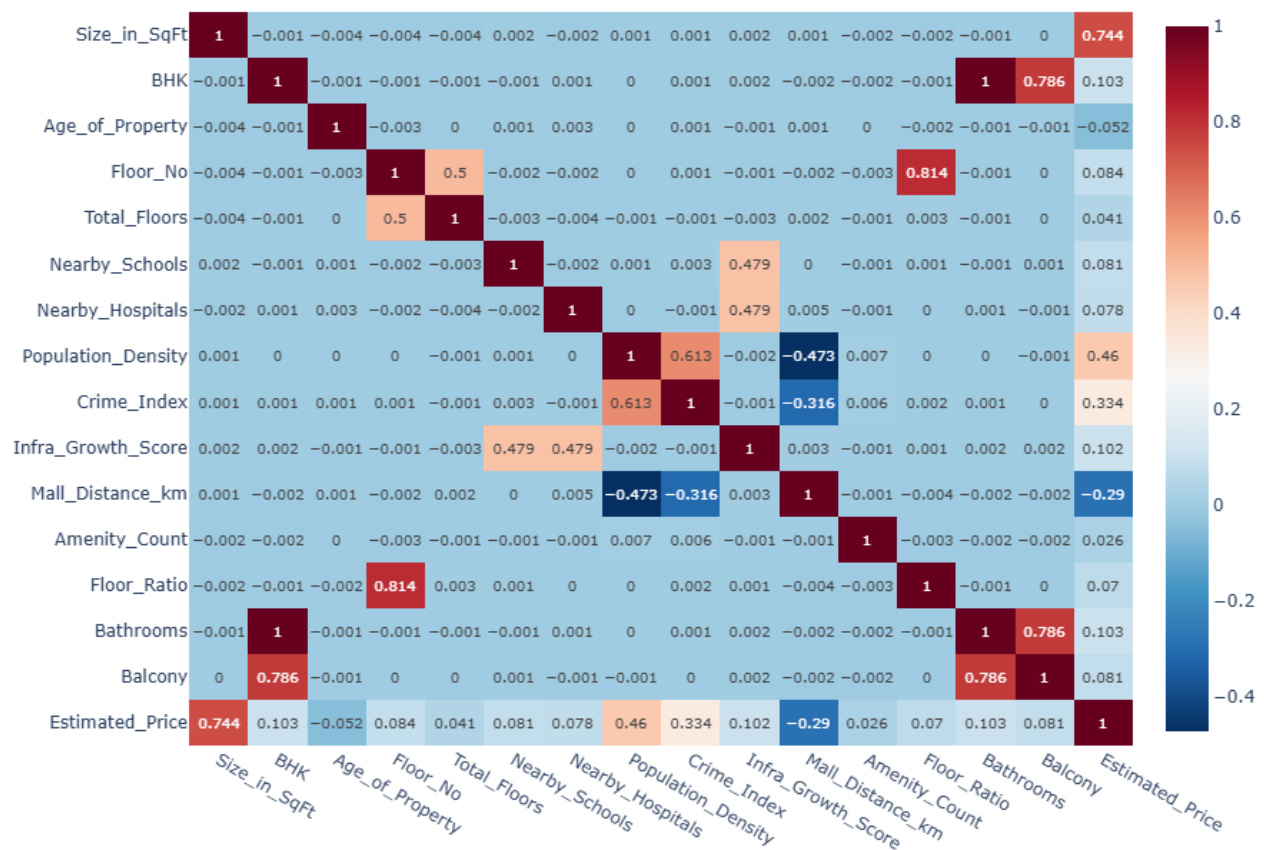


Figure 5: Correlation heatmap matrix showing relationships between numerical variables.

Key Heatmap Insight: Interestingly, Mall_Distance_km showed a negative correlation (-0.290), indicating that properties located closer to shopping malls tend to command higher prices. Conversely, the Age_of_Property showed a slight negative correlation (-0.052), meaning newer properties are slightly more expensive.

5. Key Findings and Insights

- **Size is the Ultimate Driver:** The built-up area (Size_in_SqFt) is by far the strongest predictor of a property's price, with a 0.744 correlation.
- **The Premium of Density:** Population density plays a significant role (0.460 correlation). Cities with higher density (like Mumbai and Delhi) command significantly higher real estate prices compared to tier-2 cities.

- **Mumbai Leads the Market:** Mumbai has the highest average property price at ₹4.78 Crore (478.09 Lakhs), followed closely by New Delhi.
- **Proximity to Commercial Hubs Matters:** Properties closer to malls are valued higher, evidenced by the negative correlation between distance to malls and price.
- **Perfectly Balanced Market:** The dataset reveals an almost perfectly balanced inventory split between Under Construction (125,035) and Ready to Move (124,965) properties.
- **Furnishing does not dictate major volume:** The market is evenly split across Furnished, Semi-furnished, and Unfurnished properties (roughly 83,000 each).
- **Infrastructure Growth Impact:** The engineered Infra_Growth_Score has a positive correlation (0.102) with price, validating that better nearby schools, hospitals, and transit positively impact valuation.
- **Age Depreciation is Minimal:** Property age has a surprisingly weak negative correlation (-0.052) with price, suggesting that location and land value heavily outweigh building depreciation in the Indian market.

6. Outlier Analysis

Based on the box plots and interquartile range checks performed earlier, the dataset was largely free of extreme statistical anomalies in the core dimensions (Price and Size). The prices were naturally capped at 500 Lakhs and sizes at 5000 SqFt during data generation. However, high-variance was noted in the price per square foot dimension, reflecting the natural reality of ultra-premium locations versus suburban areas.

7. Conclusion

The Exploratory Data Analysis phase successfully mapped the contours of the dataset. We determined that Property Size, Population Density (City Tier), and Proximity to Commercial Hubs are the primary drivers of real estate valuation. The linear relationships observed here provide a strong foundation for building predictive regression algorithms in the subsequent phases.